João Eduardo Pereira dos Santos

Senior Python & AI Backend Engineer | Distributed Systems & Cloud

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Senior Software Engineer with **16+ years of experience** architecting resilient Python backend microservices, enterprise data integration pipelines, and high-performance cloud applications (AWS, Azure, GCP). Specializing in **Python (FastAPI, AsyncIO, Flask, Django)**, **RAG/LLM Agents**, and mission-critical relational databases (PostgreSQL, Oracle, SQL). Proven track record delivering enterprise-grade platforms at **Critical TechWorks (BMW Group Partner)**, **Ernst & Young**, and **UNICAMP**. Native Portuguese, verified C2 English.

PROFESSIONAL EXPERIENCE

Python Full Stack Developer – Critical TechWorks (BMW Partner) Nov 2025 – Present | Porto, Portugal

- Architect and integrate Python backend microservices, REST APIs, and automotive data solutions with growing focus on AI-enabled engineering.
- Deploy and maintain containerized workloads using Docker & Kubernetes on AWS cloud infrastructure.
- Implement automated CI/CD pipelines, regression testing suites, and agile delivery (Scrum).
- Optimize complex SQL data extraction, transformation, and enterprise system connectivity.
- Collaborate with international BMW engineering teams across Germany and Portugal delivering high-reliability services.

Python Full Stack & AI Developer – Ernst & Young (EY) Dec 2022 – Nov 2025 | Brazil / Hybrid

- Developed Python-based Retrieval-Augmented Generation (RAG) platform for semantic search and Q&A over enterprise document repositories.
- Engineered document ingestion, chunking algorithms, vector embeddings, and LLM contextual reasoning.
- Delivered large-scale SAP migration pipelines, IBM DataStage ETL workflows, and SQL validation automation.
- Built analytics dashboards and secure REST API services connecting legacy enterprise datastores.
- Authored complex SQL transformations and automated data pipelines for multi-terabyte corporate datasets.

Senior Systems Engineer & Python Developer – UNICAMP Apr 2008 – Dec 2022 (14+ yrs) | Campinas, Brazil

- Led full-stack engineering for institutional platforms using Python (Django, Flask), React, and REST APIs.
- Architected and delivered custom administrative and research portal backends serving 40,000+ university users.
- Modeled, tuned, and maintained mission-critical databases (PostgreSQL, MySQL, Oracle) handling campus-wide traffic.
- Spearheaded digital transformation, DevOps automation, Linux server administration, and BI reporting pipelines.
- Directed database migrations, performance tuning, and automated backups across heterogeneous clusters.

Earlier Career: *Digital Games Programming Professor* at SESC Santana (2016–2019; Python, Lua) • *Software Analyst* at Mercer Brasil (2005–2008; SQL Server, ETL, database integration) • *Technical Support* at Atento (2001–2002).

TECHNICAL EXPERTISE

Backend & Languages

Python (16+ yrs) | FastAPI | AsyncIO | Flask | Django | REST APIs | JavaScript / React | TypeScript | Node.js

AI & Data Engineering

LLMs & Agents | RAG Architectures | Vector DBs (Chroma, PGVector) | Embeddings | Prompt Eng | IBM DataStage | SAP Migration | ETL / ELT

Databases & SQL

PostgreSQL | Oracle DB | MySQL | SQL Server | Query Optimization

Cloud & DevOps

AWS | Azure | GCP | Docker | Kubernetes | Jenkins | Git / CI/CD | Linux Admin

Architecture & Practices

Clean Architecture | Microservices | Event-Driven | TDD / Unit Testing | Agile / Scrum

EDUCATION & TRAINING

B.S. Computer Engineering
UNIVESP (Brazil)
2019 – Present

Data Processing Technologist
FATEC-SP (Brazil)
2003 – 2007

Building Web Apps in Django
University of Michigan

Full Stack Developer Accredited
PUC-Rio

LANGUAGES

English	C2 Proficient / Mastery
Portuguese	Native